TA14 AUTHORITY GOVERNANCE INSTITUTION · TA-14 EXCHANGE · TECHNICAL EXAMINATION v1 · PROPOSED, NOT VALIDATED

WHEN THE ATTACKER BYPASSES GOVERNANCE

Refrigeration Cybersecurity, Physical Drift, and Admissible Evidence

Revision 1.1: adds the emergency-response pathway and extension scenario E1 (section 9A); the origin statement is revised; scenarios S1–S8 and their results are unchanged.

SIMULATION — NOT PHYSICAL-SYSTEM VALIDATION. Every result in this document comes from a deterministic software simulation. No physical refrigeration system, controller, network or sensor was tested.

1. Executive summary

This examination asks whether an unauthorized change to refrigeration equipment can be **prevented** when it passes through TA-14's Admissible Execution Architecture (AEA), **detected** by independent observation when it bypasses AEA, and **preserved** as an inspectable incident record. Central principle: *an attacker may bypass the governance boundary without automatically escaping independent observation and evidence preservation*. That is a conditional capability, not a guarantee.

Eight adversarial scenarios were run. Across them, 27 examined claims passed, **3 failed** and 0 were not demonstrated:

- **Prevention** held for commands submitted through AEA (S1, S7) and **failed** for commands that bypassed it (S2, S3). AEA cannot refuse a command it never sees.
- **Detection** by independent sensors succeeded in S2–S5 (S2 after 10 min, S3 after 35 min, S4 after 35 min, S5 after 35 min), including against a controller that falsified its reports (S4) and a suppressed alarm (S5). It **failed** in S6, when the independent temperature evidence was lost; the gap was disclosed, and the corrective action was held for lack of admissible evidence.
- **Evidence preservation**: all eight hash-chained records verify; five tamper types are detected; no authorization was fabricated for bypassed commands; no cause was attributed.
- **Emergency pathway** (extension E1, section 9A): with independent evidence lost, a pre-authorized protective measure that only re-asserts a sealed-baseline value was allowed; consequential commands are never available on that path; ordinary governance still held. Detection and outcome verification remained NOT DEMONSTRATED. 22 adversarial tests show the pathway cannot be used to bypass ordinary governance. S6 is preserved as a failure case.
- **Recovery** (S8): a routine command was held during the open incident, a corrective authority revoked between commit and execution was refused at the execution boundary, and a fresh authority restored the baseline, verified afterwards by independent observation.

Capability	S1	S2	S3	S4	S5	S6	S7	S8
PREVENTION	PASS	FAIL	FAIL	·	·	·	PASS	·
NO_FALSE_ALERT	PASS	·	·	·	·	·	PASS	·

EVIDENCE_PRESERVATION	PASS	PASS	PASS	PASS	PASS	PASS	PASS	PASS
DETECTION	·	PASS	PASS	PASS	PASS	FAIL	·	·
NO_FABRICATED_AUTHORIZATION	·	PASS	PASS	·	·	·	·	·
ATTRIBUTION_NOT_ASSERTED	·	PASS	·	·	·	·	·	·
DETECTION_INDEPENDENT_OF_CONTROLLER_REPORTS	·	·	·	PASS	·	·	·	·
ALERT_INDEPENDENT_OF_CONTROLLER_ALARM	·	·	·	·	PASS	·	·	·
EVIDENCE_GAP_DISCLOSED	·	·	·	·	·	PASS	·	·
NO_EXECUTION_WITHOUT_ADMISSIBLE_EVIDENCE	·	·	·	·	·	PASS	·	·
ROUTINE_COMMAND_HELD_DURING_INCIDENT	·	·	·	·	·	·	·	PASS
RUNTIME_REVALIDATION	·	·	·	·	·	·	·	PASS
RECOVERY_AUTHORIZED	·	·	·	·	·	·	·	PASS
POST_INTERVENTION_VERIFICATION	·	·	·	·	·	·	·	PASS

2. Origin of the challenge

This is an independent TA-14 technical examination. It examines a question that arises wherever refrigeration equipment is networked: what happens when an unauthorized actor changes equipment behaviour outside its intended governance boundary? No external party has reviewed, participated in, or endorsed this examination or its conclusions.

3. System architecture

Component	Status in this examination	Role
Thermal plant model	Simulation (illustrative, unvalidated)	Illustrative lumped two-node model; parameters chosen for teaching, not measured. Two nodes (cabinet air, product); τ cool 25 min, τ leak 240 min, τ defrost 40 min, τ product 300 min; room 24 °C.
Controller	Simulation	Thermostat and defrost schedule; can be written to directly (bypass), can falsify its reports and can suppress its alarm.
AEA execution gate	**Production GCEA module, unchanged**, plus simulation actor and bounds checks that can only tighten a determination	Evidence, authority, standing, material change, commit, and revalidation at execution (ALLOW / HOLD / DENY / ESCALATE).
EIN	Simulation	Two product-simulant probes, a cabinet air probe, and heater and compressor current clamps on separate loggers. It never reads ground truth or trusts the controller.
EIR	Simulation	Append-only SHA-256 hash chain anchored to the sealed baseline, with explicit evidence gaps.
Response	Simulation (simulated people)	Notification, acknowledgement, investigation, re-authorization, revalidated corrective action, post-intervention observation.

4. Threat model and trust boundaries

Modelled attacker capabilities: submitting commands through the governed route without authority; replaying a genuine expired authority; writing directly to the controller (defrost, setpoint); making the controller report false temperatures; suppressing the controller alarm. **Independently modelled failure:** loss or sticking of the independent temperature probes.

Not modelled: compromise of the independent probes, their loggers or the recorder; theft of a current, unexpired authority together with the holder's identity; physical tampering; network-level attacks; real controller protocols.

Trust boundaries: (1) the AEA gate governs only commands that pass through it; (2) the controller's self-reports are recorded but treated as untrusted (*independentOfController: false*); (3) the EIN's independence from the controller is an *assumption* of the simulation, which real sites must establish; (4) the simulation's ground truth exists only for examination and is never supplied to the EIN, EIR or gate.

5. Authorized baseline

Item	Value
Baseline	RCB-BL-1.0, Simulated reach-in display freezer RCB-FREEZER-01, controller CTRL-SIM-01
Setpoint / differential	-22 °C / 2 K
Approved air band	-25 to -19 °C
Product limit / exception	-18 °C / -15 °C (configured; Configured for this simulation. -18 °C is a commonly cited frozen-storage temperature; site limits must come from the operator's food-safety plan and equipment documentation.)
Defrost	02:00, 08:00, 14:00, 20:00 for 20 min (max 30; recovery allowance 40 min)
Operating limits	Setpoint -25 to -18 °C; defrost ≤ 30 min
Actors	FM-01 (FACILITY_MANAGER, CURRENT); TECH-01 (REFRIGERATION_TECHNICIAN, CURRENT)
Evidence sources	P1: product-simulant probe (glycol bottle) on EIN-LOGGER-A; P2: product-simulant probe (glycol bottle) on EIN-LOGGER-B; AIR: cabinet air probe on EIN-LOGGER-A; HEATER_CT: defrost heater current clamp on EIN-LOGGER-B; COMPRESSOR_CT: compressor current clamp on EIN-LOGGER-B
Seal (SHA-256)	`797f9d8dfa5f5d213116836aa8a241acb651dfa3eb373bd8976e7f4b96e411f4`

6. AEA execution-boundary behavior

Scenario	Command	Commit	Execution (revalidated)	Reasons
S1	Unauthorized 6-hour defrost via AEA	DENY	not attempted	`SIM_ACTOR_STANDING_NOT_ESTABLISHED, SIM_COMMAND_OUTSIDE_AUTHORIZED_LIMITS, AUTHORITY_ABSENT, COMMIT_BOUNDARY_REFUSED`
S2	Corrective action: restore the authorized baseline	ALLOW	ALLOW	`PRESENT_STANDING_ESTABLISHED, COMMIT_BOUNDARY_ESTABLISHED, PRESENT_STANDING_ESTABLISHED, EXECUTION_BOUNDARY_PERMITTED`

S3	Corrective action: restore the authorized baseline	ALLOW	ALLOW	`PRESENT_STANDING_ESTABLISHED, COMMIT_BOUNDARY_ESTABLISHED, PRESENT_STANDING_ESTABLISHED, EXECUTION_BOUNDARY_PERMITTED`
S4	Corrective action: restore the authorized baseline	ALLOW	ALLOW	`PRESENT_STANDING_ESTABLISHED, COMMIT_BOUNDARY_ESTABLISHED, PRESENT_STANDING_ESTABLISHED, EXECUTION_BOUNDARY_PERMITTED`
S5	Corrective action: restore the authorized baseline	ALLOW	ALLOW	`PRESENT_STANDING_ESTABLISHED, COMMIT_BOUNDARY_ESTABLISHED, PRESENT_STANDING_ESTABLISHED, EXECUTION_BOUNDARY_PERMITTED`
S6	Corrective action: restore the authorized baseline	HOLD	not attempted	`CONTINUITY_UNSUPPORTED, ADMISSIBILITY_UNSUPPORTED, COMMIT_BOUNDARY_REFUSED`
S7	Expired authority replayed by an unrecognized actor	DENY	not attempted	`SIM_ACTOR_STANDING_NOT_ESTABLISHED, SIM_ACTOR_IS_NOT_GRANT_HOLDER, AUTHORITY_EXPIRED, COMMIT_BOUNDARY_REFUSED`
S7	Expired authority replayed by its original holder	DENY	not attempted	`AUTHORITY_EXPIRED, COMMIT_BOUNDARY_REFUSED`
S8	Routine defrost attempted during the open incident (current grant)	HOLD	not attempted	`MATERIAL_CHANGE_REQUIRES_REAUTHORIZATION, COMMIT_BOUNDARY_REFUSED`
S8	Corrective action: restore the authorized baseline	ALLOW	DENY	`PRESENT_STANDING_ESTABLISHED, COMMIT_BOUNDARY_ESTABLISHED, AUTHORITY_REVOKED, EXECUTION_BOUNDARY_REFUSED`
S8	Corrective action (second authority)	ALLOW	ALLOW	`PRESENT_STANDING_ESTABLISHED, COMMIT_BOUNDARY_ESTABLISHED, PRESENT_STANDING_ESTABLISHED, EXECUTION_BOUNDARY_PERMITTED`

Bypassed commands (S2–S6, S8) produced no AEA entry. The record states that an observed actuation has no authorization record (investigation finding *NO_AUTHORIZATION_RECORD_FOR_OBSERVED_STATE_CHANGE*).

7. Independent physical observation

Statuses: NORMAL · WATCH (product above -18 °C, or air outside the band for ≥ 10 min outside an authorized defrost) · EXCEPTION (product above -15 °C) · FAULT (heater current outside every authorized defrost window; cooling not responding; controller report disagreeing with the independent air probe by > 3 K for 3 samples) · EVIDENCE_INSUFFICIENT (product or air state cannot be established: missing, stuck for 6 samples, or disagreeing probes). Every finding is CAUSE_UNDETERMINED, with candidate causes listed.

Scenario	Change at 03:00 (ground truth)	First independent deviation	Latency	Max EIN status
S2	Direct controller write: defrost for 360 min	03:10	10 min	FAULT
S3	Direct controller write: setpoint +2 °C	03:35	35 min	FAULT
S4	Direct controller write: setpoint +2 °C and falsified reporting	03:35	35 min	FAULT
S5	Direct controller write: setpoint +2 °C and alarm suppressed	03:35	35 min	FAULT
S6	Direct controller write: setpoint +2 °C	**not detected**	—	EVIDENCE_INSUFFICIENT

On normal days with four authorized defrosts (S1, S7) the EIN raised no alert.

8. EIN/EIR evidence preservation

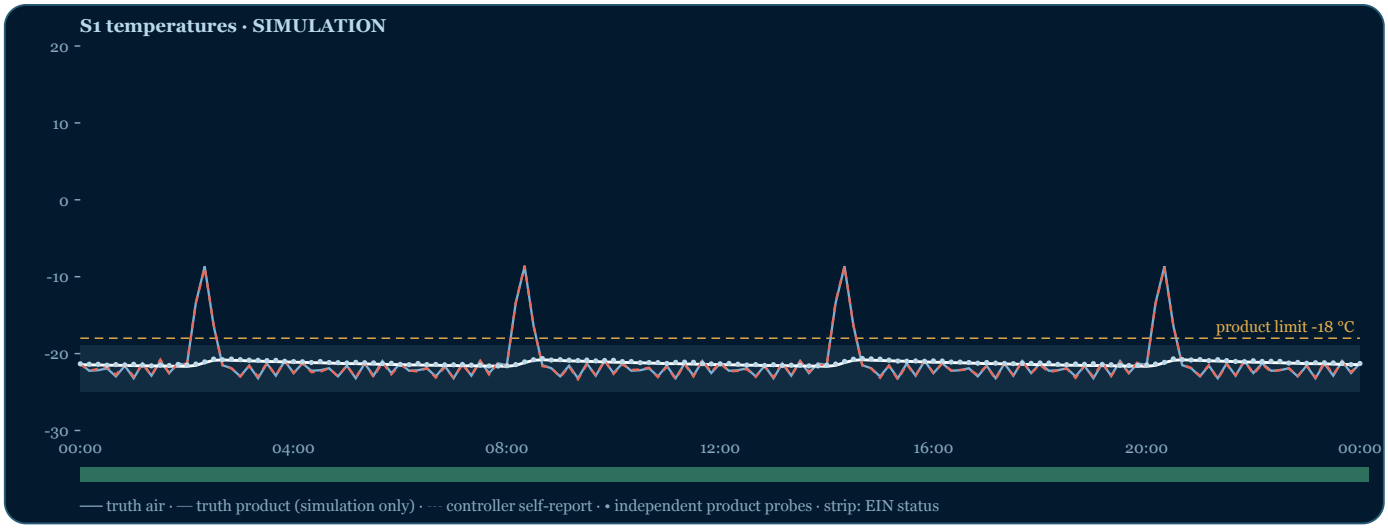
Each entry records sequence, recorder time, observation time, source (EIN, AEA_GATE, CONTROLLER_REPORTED, HUMAN, RECORDER), independence from the controller, payload, the previous entry's hash and its own SHA-256 over canonical JSON. The sealed baseline is stored separately and anchors the chain (genesis = seal). Verification recomputes every hash and link, sequence continuity, monotonic recorder time and the chain head, in Node or in a browser via WebCrypto (same code).

Limitations: entries are not signed and carry no trusted third-party time-stamp. A valid chain shows the export was not altered; it does not show the sensors were right, and an existing record is not automatically admissible.

9. The eight adversarial scenarios

S1 · Unauthorized defrost command submitted through AEA

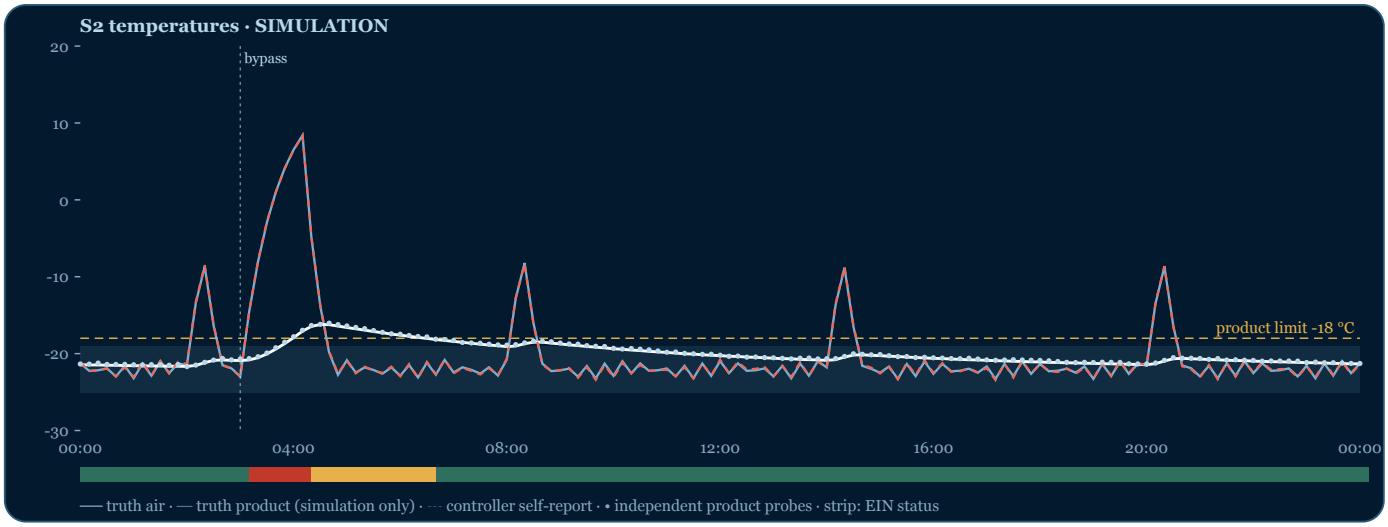
Attempted	An unrecognized remote actor submits a 6-hour defrost through the governed command route, without any authority grant.
What AEA observed	• 03:00 · Unauthorized 6-hour defrost via AEA (actor UNKNOWN-REMOTE-01): **DENY** · `SIM_ACTOR_STANDING_NOT_ESTABLISHED, SIM_COMMAND_OUTSIDE_AUTHORIZED_LIMITS, AUTHORITY_ABSENT, COMMIT_BOUNDARY_REFUSED`
What EIN independently observed	NORMAL throughout.
What EIR preserved	581 hash-chained entries (BASELINE_REFERENCE 1, CONTROLLER_REPORT 289, EIN_OBSERVATION 289, AEA_COMMAND_SUBMITTED 1, OUTCOME_RECORDED 1); verification VALID; 0 evidence gaps; chain head `1d088c9b71e22779411564fd78828456…`
Alert	No EIN alert. No controller alarm.
Physical change (simulation ground truth)	No deviation beyond authorized defrosts (product max -20.81 °C).
Outcome	No physical deviation. Product stayed at or below -18 °C (max -20.81 °C).
Claims	PASS **PREVENTION**: CMD-S1-DEFROST-360: DENY (SIM_ACTOR_STANDING_NOT_ESTABLISHED, SIM_COMMAND_OUTSIDE_AUTHORIZED_LIMITS, AUTHORITY_ABSENT, COMMIT_BOUNDARY_REFUSED); executed=false PASS **NO_FALSE_ALERT**: Four authorized defrost cycles ran; the EIN raised no alert. PASS **EVIDENCE_PRESERVATION**: Chain valid: true (581 entries); sealed baseline unchanged: true; missing entry types: none.
What remains unproven	• That every real command path passes through the gate. A bypass path is examined in S2. • Physical-system behaviour: the thermal model is illustrative and unvalidated. • Real controller protocols, network paths and attacker capabilities were not tested. • Detection depends on independent sensors existing, being installed correctly and staying independent of the compromised path.



S2 · Unauthorized defrost command bypassing AEA

Attempted	An attacker writes a 6-hour defrost directly to the controller, outside the governed route.
What AEA observed	• 03:55 · Corrective action: restore the authorized baseline (actor TECH-01): **ALLOW** → execution **ALLOW** · `PRESENT_STANDING_ESTABLISHED, COMMIT_BOUNDARY_ESTABLISHED, PRESENT_STANDING_ESTABLISHED, EXECUTION_BOUNDARY_PERMITTED`
What EIN independently observed	03:10 FAULT → 04:15 WATCH → 06:40 NORMAL First deviation finding at 03:10.
What EIR preserved	591 hash-chained entries (BASELINE_REFERENCE 1, CONTROLLER_REPORT 289, EIN_OBSERVATION 289, EIN_STATUS_CHANGED 3, EIN_ALERT 1, NOTIFICATION_SENT 1, HUMAN_ACKNOWLEDGED 1, INVESTIGATION_RECORDED 1, CONTROLLER_ALARM_REPORTED 1, AUTHORITY_ISSUED 1, AEA_COMMAND_SUBMITTED 1, AEA_COMMAND_EXECUTED 1, OUTCOME_RECORDED 1); verification VALID; 0 evidence gaps; chain head `569a63efd8825e5fbcb82dcf47bf3aaf…`
Alert	EIN alerts: FAULT at 03:10 Controller self-reported alarm at 03:44.
Physical change (simulation ground truth)	Yes. Product peaked at -16.2 °C; 149 min above -18 °C; -21.33 °C at the end.
Outcome	Baseline restored under fresh authority at minute 250. Simulated product peaked at -16.2 °C and spent 149 min above -18 °C; at the end it was -21.33 °C. Product disposition is not decided by TA-14.
Claims	FAIL **PREVENTION**: The command bypassed the governed route and changed controller behaviour. AEA cannot prevent a command that never reaches it. PASS **DETECTION**: Ground-truth deviation from minute 181; independent EIN flagged a deviation at minute 190 (10 min after the change). PASS **NO_FABRICATED_AUTHORIZATION**: No AEA determination or execution record exists for the bypassed command; the record states the absence instead. PASS **ATTRIBUTION_NOT_ASSERTED**: Attributions recorded: CAUSE_UNDETERMINED. PASS **EVIDENCE_PRESERVATION**: Chain valid: true (591 entries); sealed baseline unchanged: true; missing entry types: none.

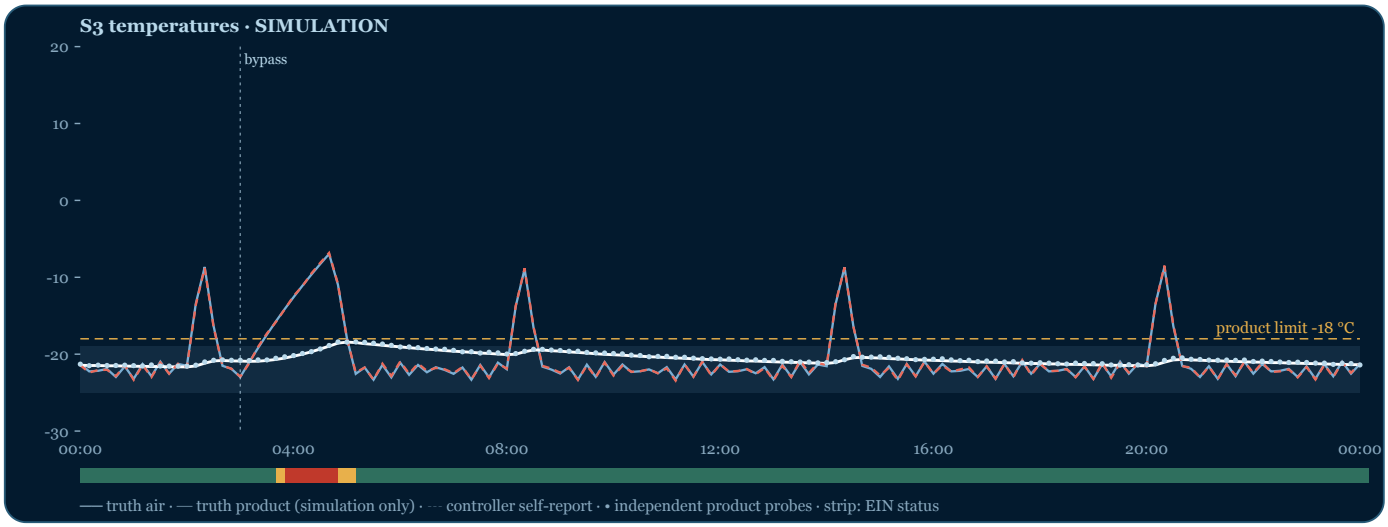
What remains unproven	• That the cause was malicious: the record shows an actuation with no authorization record, not who or why. • Physical-system behaviour: the thermal model is illustrative and unvalidated. • Real controller protocols, network paths and attacker capabilities were not tested. • Detection depends on independent sensors existing, being installed correctly and staying independent of the compromised path.



S3 · Unauthorized temperature setpoint change

Attempted	An attacker changes the controller setpoint from –22 °C to +2 °C outside the governed route.
What AEA observed	• 04:30 · Corrective action: restore the authorized baseline (actor TECH-01): **ALLOW** → execution **ALLOW** · `PRESENT_STANDING_ESTABLISHED, COMMIT_BOUNDARY_ESTABLISHED, PRESENT_STANDING_ESTABLISHED, EXECUTION_BOUNDARY_PERMITTED`
What EIN independently observed	03:35 WATCH → 03:45 FAULT → 04:50 WATCH → 05:05 NORMAL First deviation finding at 03:35.
What EIR preserved	593 hash-chained entries (BASELINE_REFERENCE 1, CONTROLLER_REPORT 289, EIN_OBSERVATION 289, EIN_STATUS_CHANGED 4, EIN_ALERT 2, NOTIFICATION_SENT 1, HUMAN_ACKNOWLEDGED 1, INVESTIGATION_RECORDED 1, AUTHORITY_ISSUED 1, AEA_COMMAND_SUBMITTED 1, CONTROLLER_ALARM_REPORTED 1, AEA_COMMAND_EXECUTED 1, OUTCOME_RECORDED 1); verification VALID; 0 evidence gaps; chain head `a6ece877e4a69442e7b6a50970087705…`
Alert	EIN alerts: WATCH at 03:35, FAULT at 03:45 Controller self-reported alarm at 04:33.
Physical change (simulation ground truth)	Yes. Product peaked at -18.43 °C; 0 min above -18 °C; -21.37 °C at the end.
Outcome	Baseline restored under fresh authority at minute 285. Simulated product peaked at -18.43 °C and spent 0 min above -18 °C; at the end it was -21.37 °C. Product disposition is not decided by TA-14.
Claims	FAIL **PREVENTION**: The command bypassed the governed route and changed controller behaviour. AEA cannot prevent a command that never reaches it. PASS **DETECTION**: Ground-truth deviation from minute 212; independent EIN flagged a deviation at minute 215 (35 min after the change). PASS **NO_FABRICATED_AUTHORIZATION**: No AEA determination or execution record exists for the bypassed command; the record states the absence instead. PASS **EVIDENCE_PRESERVATION**: Chain valid: true (593 entries); sealed baseline unchanged: true; missing entry types: none.

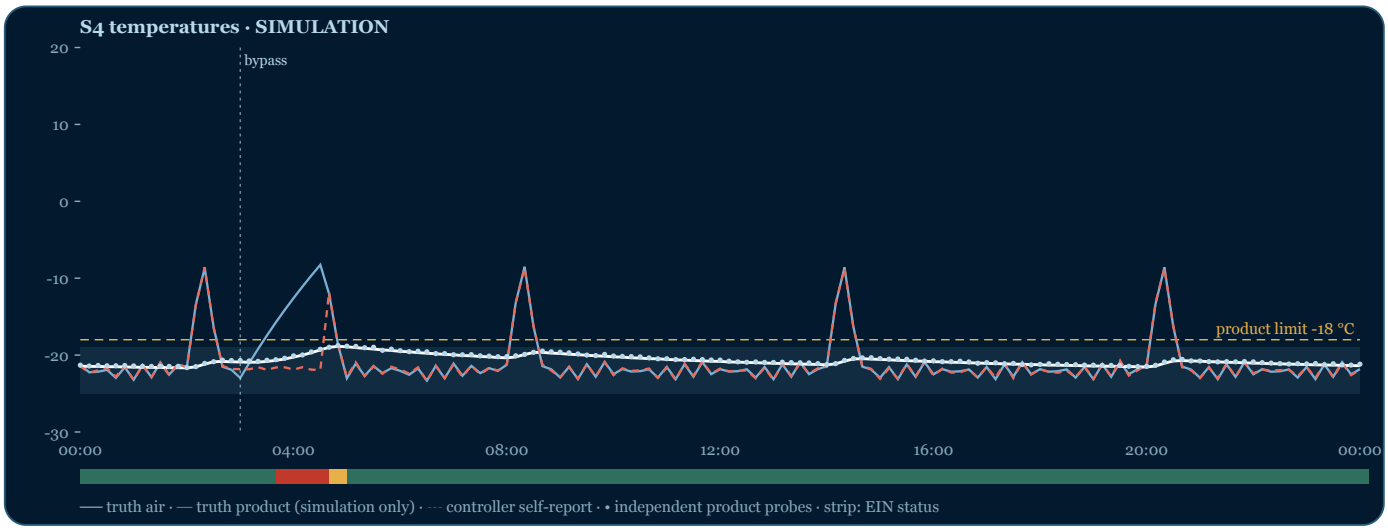
What remains unproven	• Detection speed for slower or subtler setpoint drift (e.g. –22 → –19 °C) was not examined. • Physical-system behaviour: the thermal model is illustrative and unvalidated. • Real controller protocols, network paths and attacker capabilities were not tested. • Detection depends on independent sensors existing, being installed correctly and staying independent of the compromised path.



S4 · Compromised controller reporting false temperatures

Attempted	An attacker raises the setpoint to +2 °C and makes the controller report a steady, plausible –21.8 °C.
What AEA observed	• 04:20 · Corrective action: restore the authorized baseline (actor TECH-01): **ALLOW** → execution **ALLOW** · `PRESENT_STANDING_ESTABLISHED, COMMIT_BOUNDARY_ESTABLISHED, PRESENT_STANDING_ESTABLISHED, EXECUTION_BOUNDARY_PERMITTED`
What EIN independently observed	03:35 FAULT → 04:40 WATCH → 04:55 NORMAL First deviation finding at 03:35.
What EIR preserved	590 hash-chained entries (BASELINE_REFERENCE 1, CONTROLLER_REPORT 289, EIN_OBSERVATION 289, EIN_STATUS_CHANGED 3, EIN_ALERT 1, NOTIFICATION_SENT 1, HUMAN_ACKNOWLEDGED 1, INVESTIGATION_RECORDED 1, AUTHORITY_ISSUED 1, AEA_COMMAND_SUBMITTED 1, AEA_COMMAND_EXECUTED 1, OUTCOME_RECORDED 1); verification VALID; 0 evidence gaps; chain head `3ad8e79d2c1d8cd151618a6f2029a6ec…`
Alert	EIN alerts: FAULT at 03:35 No controller alarm.
Physical change (simulation ground truth)	Yes. Product peaked at -18.86 °C; 0 min above -18 °C; -21.38 °C at the end.
Outcome	Baseline restored under fresh authority at minute 275. Simulated product peaked at -18.86 °C and spent 0 min above -18 °C; at the end it was -21.38 °C. Product disposition is not decided by TA-14.
Claims	PASS **DETECTION**: Ground-truth deviation from minute 212; independent EIN flagged a deviation at minute 215 (35 min after the change). PASS **DETECTION_INDEPENDENT_OF_CONTROLLER_REPORTS**: Controller reports during the incident all inside the approved band: true; independent EIN deviation at minute 215. PASS **EVIDENCE_PRESERVATION**: Chain valid: true (590 entries); sealed baseline unchanged: true; missing entry types: none.
What remains unproven	• An attacker who also controls the independent probes or their logger was not modelled. • Physical-system behaviour: the thermal model is illustrative and unvalidated. • Real controller protocols, network paths and attacker capabilities were not tested.

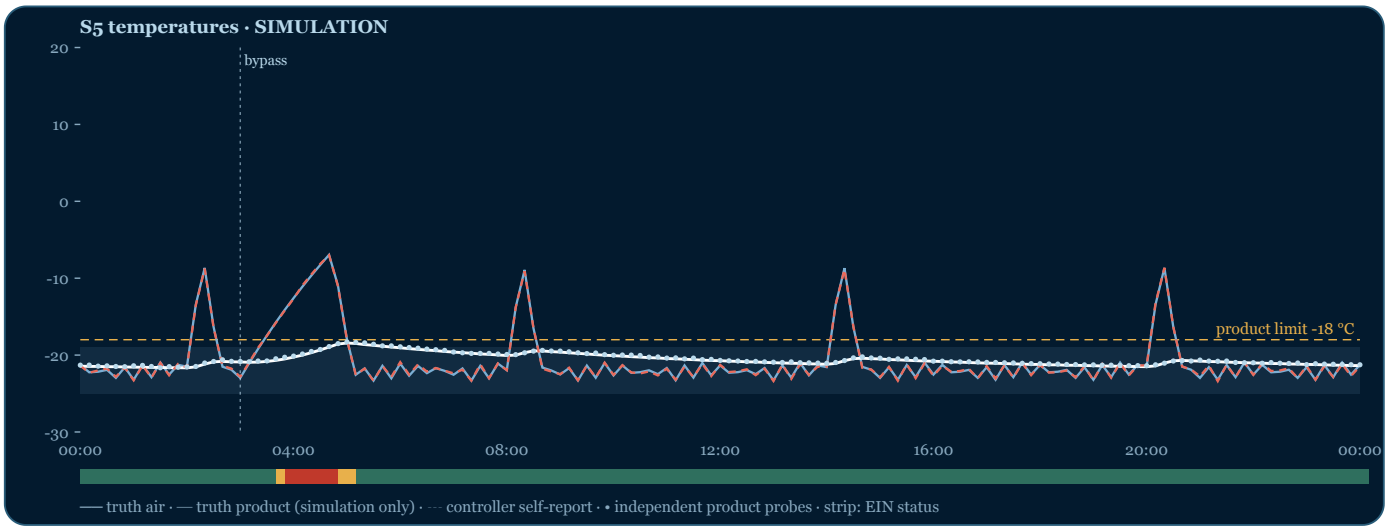
	• Detection depends on independent sensors existing, being installed correctly and staying independent of the compromised path.



S5 · Suppressed controller alarm

Attempted	An attacker raises the setpoint to +2 °C and disables the controller's own high-temperature alarm.
What AEA observed	• 04:30 · Corrective action: restore the authorized baseline (actor TECH-01): **ALLOW** → execution **ALLOW** · `PRESENT_STANDING_ESTABLISHED, COMMIT_BOUNDARY_ESTABLISHED, PRESENT_STANDING_ESTABLISHED, EXECUTION_BOUNDARY_PERMITTED`
What EIN independently observed	03:35 WATCH → 03:45 FAULT → 04:50 WATCH → 05:05 NORMAL First deviation finding at 03:35.
What EIR preserved	593 hash-chained entries (BASELINE_REFERENCE 1, CONTROLLER_REPORT 289, EIN_OBSERVATION 289, EIN_STATUS_CHANGED 4, EIN_ALERT 2, NOTIFICATION_SENT 1, HUMAN_ACKNOWLEDGED 1, INVESTIGATION_RECORDED 1, AUTHORITY_ISSUED 1, AEA_COMMAND_SUBMITTED 1, AEA_COMMAND_EXECUTED 1, CONTROLLER_ALARM_REPORTED 1, OUTCOME_RECORDED 1); verification VALID; 0 evidence gaps; chain head `7e89ea2dd9ef95b5929b171f63cc0204…`
Alert	EIN alerts: WATCH at 03:35, FAULT at 03:45 Controller self-reported alarm at 04:45. The controller alarm condition was met while the alarm was suppressed.
Physical change (simulation ground truth)	Yes. Product peaked at -18.43 °C; 0 min above -18 °C; -21.37 °C at the end.
Outcome	Baseline restored under fresh authority at minute 285. Simulated product peaked at -18.43 °C and spent 0 min above -18 °C; at the end it was -21.37 °C. Product disposition is not decided by TA-14.
Claims	PASS **DETECTION**: Ground-truth deviation from minute 212; independent EIN flagged a deviation at minute 215 (35 min after the change). PASS **ALERT_INDEPENDENT_OF_CONTROLLER_ALARM**: While suppressed (until minute 285) the controller alarm condition was met: true; controller alarms raised: 0; first independent EIN alert at minute 215. The controller alarm fired at minute 285, only after the corrective action re-enabled it. PASS **EVIDENCE_PRESERVATION**: Chain valid: true (593 entries); sealed baseline unchanged: true; missing entry types: none.
What remains unproven	• Whether a real notification path (paging, SMS, staff response) would deliver the independent alert. • Physical-system behaviour: the thermal model is illustrative and unvalidated. • Real controller protocols, network paths and attacker capabilities were not tested.

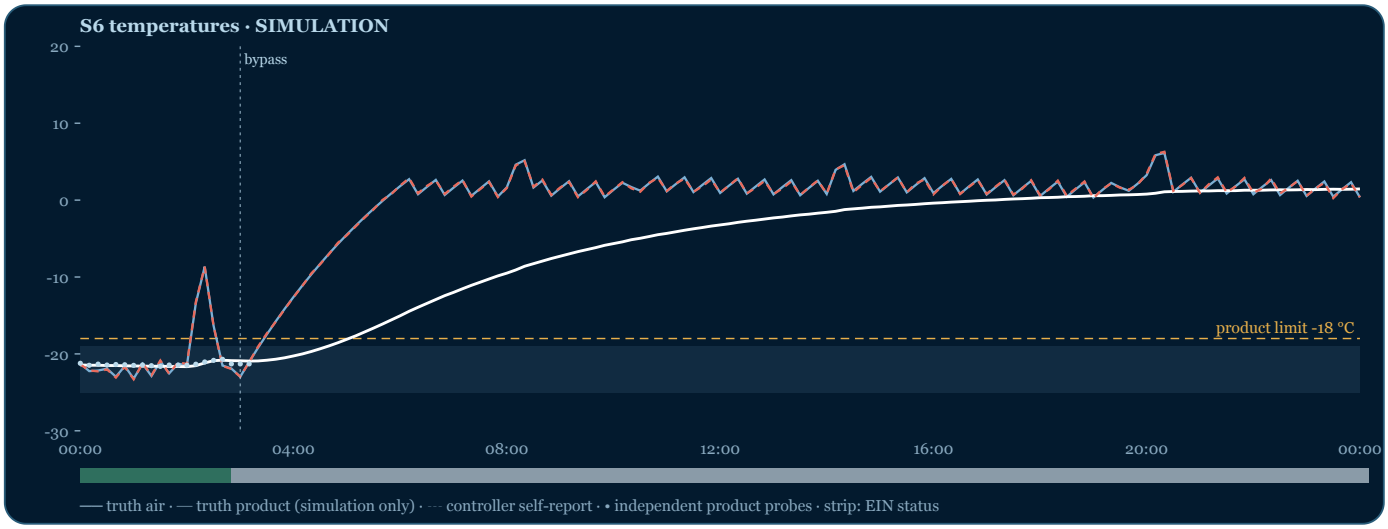
	• Detection depends on independent sensors existing, being installed correctly and staying independent of the compromised path.



S6 · Independent sensor failure or unavailable evidence

Attempted	An attacker raises the setpoint to +2 °C while the independent temperature evidence fails: P1 sticks, P2 and the air probe stop reporting.
What AEA observed	• 03:35 · Corrective action: restore the authorized baseline (actor TECH-01): **HOLD** · `CONTINUITY_UNSUPPORTED, ADMISSIBILITY_UNSUPPORTED, COMMIT_BOUNDARY_REFUSED`
What EIN independently observed	02:50 EVIDENCE_INSUFFICIENT
What EIR preserved	591 hash-chained entries (BASELINE_REFERENCE 1, CONTROLLER_REPORT 289, EIN_OBSERVATION 289, EVIDENCE_GAP_OPENED 3, EIN_STATUS_CHANGED 1, EIN_ALERT 1, NOTIFICATION_SENT 1, HUMAN_ACKNOWLEDGED 1, INVESTIGATION_RECORDED 1, AUTHORITY_ISSUED 1, AEA_COMMAND_SUBMITTED 1, CONTROLLER_ALARM_REPORTED 1, OUTCOME_RECORDED 1); verification VALID; 3 evidence gaps; chain head `b72b60c36d55dbae9a6eb14cb2915884…`
Alert	EIN alerts: EVIDENCE_INSUFFICIENT at 02:50 Controller self-reported alarm at 04:35.
Physical change (simulation ground truth)	Yes. Product peaked at 1.46 °C; 1138 min above -18 °C; 1.46 °C at the end.
Outcome	Baseline was NOT restored within the run. Simulated product peaked at 1.46 °C and spent 1138 min above -18 °C; at the end it was 1.46 °C. Product disposition is not decided by TA-14.
Claims	FAIL **DETECTION**: Ground truth deviated from minute 212 (max product 1.46 °C) but the EIN never established a deviation from its evidence. PASS **EVIDENCE_GAP_DISCLOSED**: 3 evidence gaps opened; NORMAL ever claimed without evidence: false. PASS **NO_EXECUTION_WITHOUT_ADMISSIBLE_EVIDENCE**: CMD-S6-RESTORE-1: HOLD (CONTINUITY_UNSUPPORTED, ADMISSIBILITY_UNSUPPORTED, COMMIT_BOUNDARY_REFUSED) PASS **EVIDENCE_PRESERVATION**: Chain valid: true (591 entries); sealed baseline unchanged: true; missing entry types: none.
What remains unproven	• In this run the controller's own alarm, an untrusted channel, fired while the independent evidence was missing. The simulated architecture does not let untrusted signals authorize correction, so the corrective action was held and the product kept warming. How untrusted signals should feed an emergency path is an open design question, not a demonstrated capability.

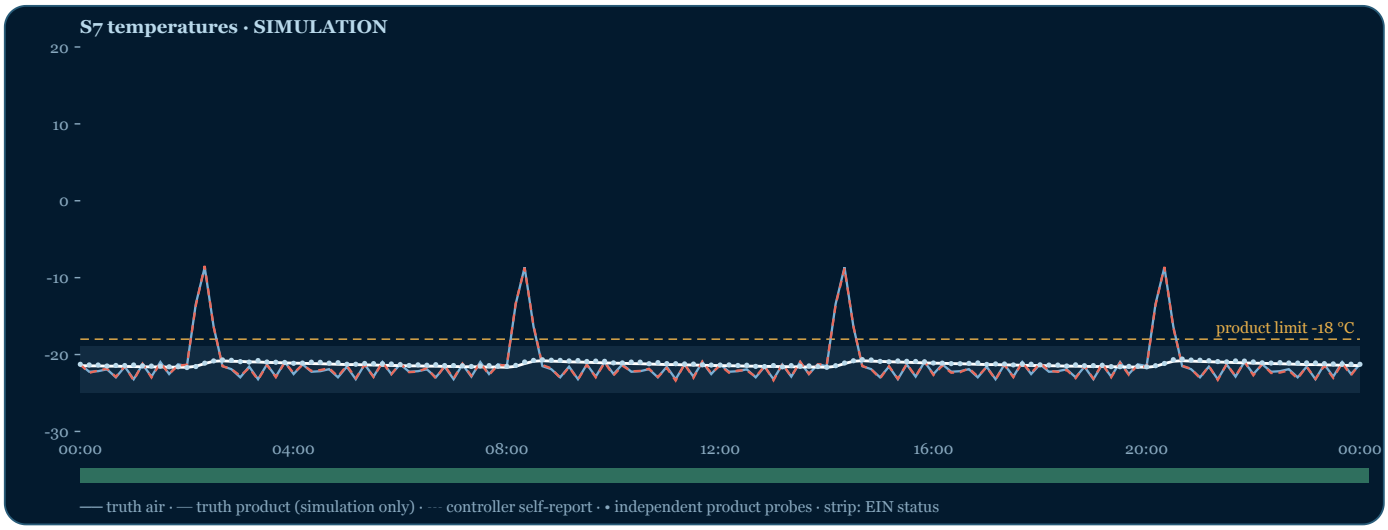
	• Any emergency procedure for acting without admissible evidence. Real sites need one; it is not modelled here. • Physical-system behaviour: the thermal model is illustrative and unvalidated. • Real controller protocols, network paths and attacker capabilities were not tested. • Detection depends on independent sensors existing, being installed correctly and staying independent of the compromised path.



S7 · Expired authorization replay

Attempted	Yesterday's genuine but expired defrost authority is replayed through the governed route: first by an unrecognized actor, then by its original holder.
What AEA observed	• 03:00 · Expired authority replayed by an unrecognized actor (actor UNKNOWN-REMOTE-01): **DENY** · `SIM_ACTOR_STANDING_NOT_ESTABLISHED, SIM_ACTOR_IS_NOT_GRANT_HOLDER, AUTHORITY_EXPIRED, COMMIT_BOUNDARY_REFUSED` • 03:05 · Expired authority replayed by its original holder (actor TECH-01): **DENY** · `AUTHORITY_EXPIRED, COMMIT_BOUNDARY_REFUSED`
What EIN independently observed	NORMAL throughout.
What EIR preserved	582 hash-chained entries (BASELINE_REFERENCE 1, CONTROLLER_REPORT 289, EIN_OBSERVATION 289, AEA_COMMAND_SUBMITTED 2, OUTCOME_RECORDED 1); verification VALID; 0 evidence gaps; chain head `d2ffe71237fa71926e564f562fa7ef21…`
Alert	No EIN alert. No controller alarm.
Physical change (simulation ground truth)	No deviation beyond authorized defrosts (product max -20.81 °C).
Outcome	No physical deviation. Product stayed at or below -18 °C (max -20.81 °C).
Claims	PASS **PREVENTION**: CMD-S7-REPLAY-ATTACKER: DENY (SIM_ACTOR_STANDING_NOT_ESTABLISHED, SIM_ACTOR_IS_NOT_GRANT_HOLDER, AUTHORITY_EXPIRED, COMMIT_BOUNDARY_REFUSED); executed=false · CMD-S7-REPLAY-HOLDER: DENY (AUTHORITY_EXPIRED, COMMIT_BOUNDARY_REFUSED); executed=false PASS **NO_FALSE_ALERT**: Four authorized defrost cycles ran; the EIN raised no alert. PASS **EVIDENCE_PRESERVATION**: Chain valid: true (582 entries); sealed baseline unchanged: true; missing entry types: none.
What remains unproven	• Theft of a current, unexpired grant together with the holder's identity was not modelled. • Physical-system behaviour: the thermal model is illustrative and unvalidated.

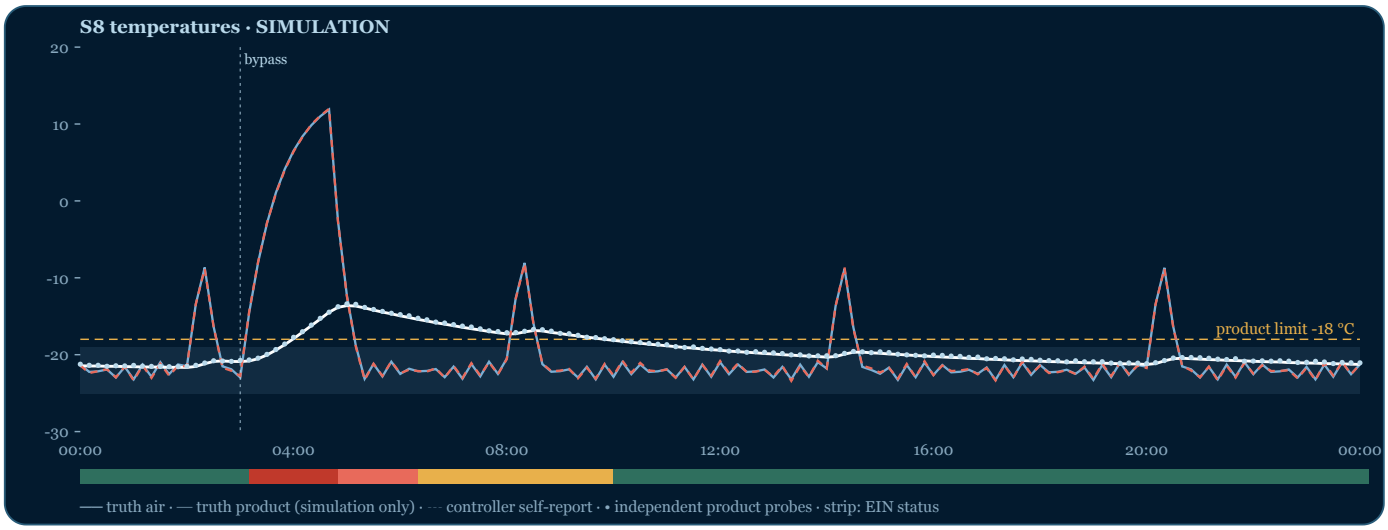
	• Real controller protocols, network paths and attacker capabilities were not tested. • Detection depends on independent sensors existing, being installed correctly and staying independent of the compromised path.



S8 · Authorized recovery with post-intervention verification

Attempted	After a bypassed 6-hour defrost, a routine command is attempted during the open incident; a corrective authority is revoked between commit and execution; a fresh authority restores the baseline; recovery is verified independently.
What AEA observed	• 03:52 · Routine defrost attempted during the open incident (current grant) (actor TECH-01): **HOLD** · `MATERIAL_CHANGE_REQUIRES_REAUTHORIZATION, COMMIT_BOUNDARY_REFUSED` • 03:55 · Corrective action: restore the authorized baseline (actor TECH-01): **ALLOW** → execution **DENY** · `PRESENT_STANDING_ESTABLISHED, COMMIT_BOUNDARY_ESTABLISHED, AUTHORITY_REVOKED, EXECUTION_BOUNDARY_REFUSED` • 04:25 · Corrective action (second authority) (actor TECH-01): **ALLOW** → execution **ALLOW** · `PRESENT_STANDING_ESTABLISHED, COMMIT_BOUNDARY_ESTABLISHED, PRESENT_STANDING_ESTABLISHED, EXECUTION_BOUNDARY_PERMITTED`
What EIN independently observed	03:10 FAULT → 04:45 EXCEPTION → 06:15 WATCH → 10:00 NORMAL First deviation finding at 03:10.
What EIR preserved	597 hash-chained entries (BASELINE_REFERENCE 1, CONTROLLER_REPORT 289, EIN_OBSERVATION 289, EIN_STATUS_CHANGED 4, EIN_ALERT 1, NOTIFICATION_SENT 1, HUMAN_ACKNOWLEDGED 1, INVESTIGATION_RECORDED 1, CONTROLLER_ALARM_REPORTED 1, AUTHORITY_ISSUED 2, AEA_COMMAND_SUBMITTED 3, AUTHORITY_REVOKED 1, AEA_EXECUTION_REFUSED 1, AEA_COMMAND_EXECUTED 1, OUTCOME_RECORDED 1); verification VALID; 0 evidence gaps; chain head `a92a0ff56e07a410f80ea7efa14a10fa…`
Alert	EIN alerts: FAULT at 03:10 Controller self-reported alarm at 03:44.
Physical change (simulation ground truth)	Yes. Product peaked at -13.62 °C; 352 min above -18 °C; -21.26 °C at the end.
Outcome	Baseline restored under fresh authority at minute 280. Simulated product peaked at -13.62 °C and spent 352 min above -18 °C; at the end it was -21.26 °C. Product disposition is not decided by TA-14.
Claims	PASS **ROUTINE_COMMAND_HELD_DURING_INCIDENT**: CMD-S8-ROUTINE-DEFROST with a current grant: HOLD (MATERIAL_CHANGE_REQUIRES_REAUTHORIZATION, COMMIT_BOUNDARY_REFUSED)

	PASS **RUNTIME_REVALIDATION**: Commit permitted: true; execution after revocation: DENY (AUTHORITY_REVOKED, EXECUTION_BOUNDARY_REFUSED) PASS **RECOVERY_AUTHORIZED**: CMD-S8-RESTORE-2 executed under ALLOW at minute 280 after revalidation. PASS **POST_INTERVENTION_VERIFICATION**: Independent EIN first NORMAL after restore at minute 600; status at end NORMAL. PASS **EVIDENCE_PRESERVATION**: Chain valid: true (597 entries); sealed baseline unchanged: true; missing entry types: none.
What remains unproven	• Product disposition: TA-14 records the temperature excursion but does not decide food safety. • That restoring controller configuration removes an attacker's access (assumed here). • Physical-system behaviour: the thermal model is illustrative and unvalidated. • Real controller protocols, network paths and attacker capabilities were not tested. • Detection depends on independent sensors existing, being installed correctly and staying independent of the compromised path.



9A. Emergency-response pathway (extension)

S6 is preserved unchanged as a documented failure: with independent evidence lost, ordinary governance held the corrective action and the product warmed. The GCEA engine is not modified to force an ALLOW. Instead, a separately authorized pathway is defined with three tiers that are never interchangeable.

Tier	Permitted	Required
E0 Emergency notification	Notify the responder and facility manager. No equipment effect.	An evidence-loss trigger; always recorded.

E1 Protective measure	**RESTORE_SEALED_SETPOINT**: Set the controller setpoint to the value in the sealed baseline. **TERMINATE_ACTIVE_DEFROST**: End any defrost in progress (cooling resumes under the thermostat). **RE_ENABLE_CONTROLLER_ALARM**: Re-enable the controller's own high-temperature alarm. Each takes no parameters and only re-asserts a sealed-baseline value.	Protocol RCB-EP-1.0 bound to the sealed baseline; a pre-issued GCEA authority scoped to route `emergency-protective-route` and the specific measure; emergency-responder standing; independent evidence lost for ≥ 15 min, computed from the EIN record (never caller-supplied); one use per measure per incident; revalidation at execution, including the condition; ordinary reauthorization required.
E2 Consequential command	Defrost, any other setpoint, disabling alarms or reporting.	**Not available on the emergency path**; ordinary governance only.

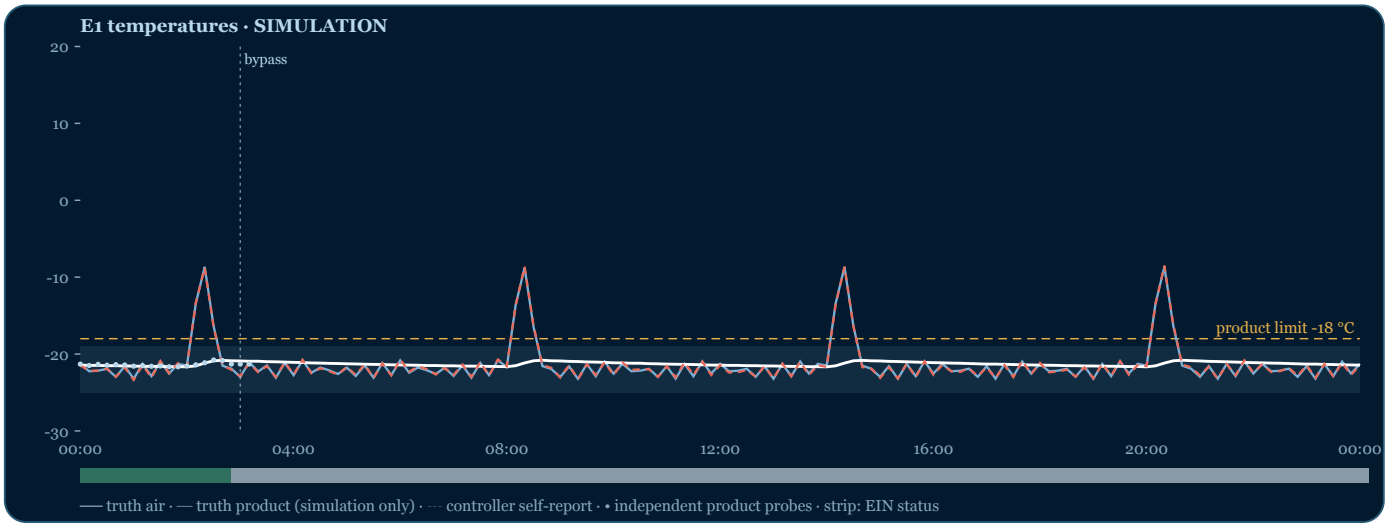
Evidence appropriate to the action. E1's evidence is the recorded, continuous EIN determination that state cannot be established, combined with the structural guarantee that the measure can only restore sealed values. GCEA receives that evidence standing honestly. When independent evidence is available, the emergency route is refused (DENY), so it cannot substitute for ordinary governance. If evidence returns before execution, execution is refused.

Adversarial tests (22, all passing): evidence available → DENY; loss not yet sustained → HOLD; actor without responder standing (even with a stolen grant) → DENY; revoked standing → DENY; consequential commands (defrost, setpoint +2 °C, disabling the alarm) → DENY; consequential actions disguised as protective measures (four variants) → DENY; smuggled parameters → DENY; no authority → refused; an ordinary grant on the emergency route → DENY (scope mismatch); one measure's grant for another → DENY; an emergency grant on the ordinary route → DENY; expired grant → DENY; revocation between commit and execution → DENY; evidence returning before execution → DENY; reuse in the same incident → DENY; protocol bound to another baseline → DENY; a caller-supplied condition is ignored.

E1 · Loss of independent evidence with the emergency protective pathway

Attempted	The S6 conditions (setpoint raised to +2 °C while the independent temperature evidence fails), with a pre-issued emergency protocol: after sustained evidence loss, the on-call responder requests the protective measure RESTORE_SEALED_SETPOINT.
What AEA observed	• 03:35 · Corrective action: restore the authorized baseline (actor TECH-01): **HOLD** · `CONTINUITY_UNSUPPORTED, ADMISSIBILITY_UNSUPPORTED, COMMIT_BOUNDARY_REFUSED`
What EIN independently observed	02:50 EVIDENCE_INSUFFICIENT
What EIR preserved	595 hash-chained entries (BASELINE_REFERENCE 1, EMERGENCY_PROTOCOL_ON_FILE 1, CONTROLLER_REPORT 289, EIN_OBSERVATION 289, EVIDENCE_GAP_OPENED 3, EIN_STATUS_CHANGED 1, EIN_ALERT 1, NOTIFICATION_SENT 1, EMERGENCY_NOTIFICATION 1, HUMAN_ACKNOWLEDGED 1, EMERGENCY_MEASURE_SUBMITTED 1, INVESTIGATION_RECORDED 1, EMERGENCY_MEASURE_EXECUTED 1, EMERGENCY_FOLLOW_UP_REQUIRED 1, AUTHORITY_ISSUED 1, AEA_COMMAND_SUBMITTED 1, OUTCOME_RECORDED 1); verification VALID; 3 evidence gaps; chain head `ab23c9a74cd55c924387539fe48d300e…`
Alert	EIN alerts: EVIDENCE_INSUFFICIENT at 02:50 No controller alarm.
Physical change (simulation ground truth)	No deviation beyond authorized defrosts (product max -20.81 °C).

Outcome	No physical deviation. Product stayed at or below -18 °C (max -20.81 °C).
Claims	PASS **EMERGENCY_PROTECTIVE_MEASURE_EXECUTED**: 190 RESTORE_SEALED_SETPOINT SUBMITTED ALLOW (PRESENT_STANDING_ESTABLISHED, COMMIT_BOUNDARY_ESTABLISHED) · 195 RESTORE_SEALED_SETPOINT EXECUTED ALLOW (PRESENT_STANDING_ESTABLISHED, EXECUTION_BOUNDARY_PERMITTED) PASS **EMERGENCY_EFFECT_LIMITED_TO_SEALED_BASELINE**: Emergency changes: [{"setpointC":-22}]; sealed values: {"setpointC":-22,"defrostUntilMinute":null,"alarmSuppressed":false}. PASS **ORDINARY_GOVERNANCE_STILL_HOLDS**: CMD-E1-RESTORE-1: HOLD (CONTINUITY_UNSUPPORTED, ADMISSIBILITY_UNSUPPORTED, COMMIT_BOUNDARY_REFUSED) NOT DEMONSTRATED **DETECTION**: No physical deviation occurred, so there was nothing to detect. NOT DEMONSTRATED **OUTCOME_VERIFIED_BY_INDEPENDENT_EVIDENCE**: Independent status at end: EVIDENCE_INSUFFICIENT. Whether the protective measure protected the product cannot be established from TA-14's evidence (simulation ground truth: product max -20.81 °C, 0 min above -18 °C). PASS **EVIDENCE_PRESERVATION**: Chain valid: true (595 entries); sealed baseline unchanged: true; missing entry types: none.
What remains unproven	• That the protective measure helped: TA-14 has no independent evidence to show it. Only the simulation ground truth shows the effect. • That a compromised controller would honour the protective write. Here it does, by assumption. • That the attacker could not simply re-apply the change after the protective measure. Repeated attacks were not modelled, and each measure may be used once per incident. • In this run the measure happened to undo the attack before the product warmed. The responder could not have known that, because the action was taken without evidence of the deviation. • That re-asserting the sealed baseline is safe for every equipment state. The protocol is designed so measures only restore sealed values, but real equipment was not examined. • Physical-system behaviour: the thermal model is illustrative and unvalidated. • Real controller protocols, network paths and attacker capabilities were not tested. • Detection depends on independent sensors existing, being installed correctly and staying independent of the compromised path.



10. Actual test results and verification procedures

Check (2026-11-03, branch feat/refrigeration-consequence-boundary from origin/main 04b58e0c)	Result

Refrigeration Consequence Boundary suite	33 passed, 0 failed (tests/acceptance/refrigeration-consequence-boundary.spec.ts)
Emergency-pathway suite	28 passed, 0 failed (tests/acceptance/refrigeration-emergency-pathway.spec.ts), of which 22 are adversarial
Full repository test suite	251 passed, 3 skipped (the gated export and report generator scripts, which run only when explicitly enabled), 0 failed; 24 test files
Type check	tsc --noEmit: 0 errors
Production build	next build: succeeded; /showrooms/refrigeration-consequence-boundary prerendered as static

Reproduce: `npx vitest run tests/acceptance/refrigeration-consequence-boundary.spec.ts`. The tests assert the verdict of every claim (including the failures), that the GCEA file equals origin/main, chain verification (server and WebCrypto), detection of five tamper types, determinism, absence of network or control endpoints, and that the committed evidence exports equal a fresh run.

Verify an export: download a record from the showroom (or `apps/web/public/showrooms/refrigeration-consequence-boundary/evidence/S1-S8.json`) and use the showroom's in-browser verifier, or call `verifyIncidentRecord` from `apps/web/lib/refrigeration-boundary/verify.ts` with any SHA-256 function. Chain heads: S1 `1d088c9b71e22779…`; S2 `569a63efd8825e5f…`; S3 `a6ece877e4a69442…`; S4 `3ad8e79d2c1d8cd1…`; S5 `7e89ea2dd9ef95b5…`; S6 `b72b60c36d55dbae…`; S7 `d2ffe71237fa7192…`; S8 `a92a0ff56e07a410….`

11. Known limitations and untested conditions

- No physical-system validation: the thermal model is illustrative, with unmeasured parameters.
- Detection depends on independent sensors that exist, are installed correctly and are genuinely independent of the compromised path. Attacks on the sensors, loggers or recorder were not modelled.
- S6: with independent evidence lost, the corrective action was held ("no admissible evidence, no admissible execution") while the controller's untrusted alarm fired, and the simulated product kept warming. An emergency procedure for acting without admissible evidence is needed in practice and is not modelled.
- The actor-standing and parameter-bounds checks are simulation additions around GCEA, not production components. EIN and EIR here are simulations, not the EIE implementation.
- Records are hash-chained but unsigned, with no trusted time-stamp. Notifications and human actions are simulated.
- Thresholds are configured for the simulation; food-safety limits and product disposition are outside TA-14's determination.
- Not examined: subtle drift (for example −22 → −19 °C), slow sensor drift, multiple simultaneous faults, theft of a current authority, real network paths and protocols.

12. Independent examination invitation

The simulation, tests, evidence exports and this report are offered for independent examination. Examiners are invited to rerun the tests, verify the published records, alter them and confirm the verifier rejects the changes, and challenge the threat model, the thresholds and the claim definitions. Findings that a claim does not hold will be recorded as findings, not corrected silently.